

Einstein–Maxwell Waves inside Weyl–Maxwell Gravity on a Compact Product: Exact Linear Phase-Space Inclusion and the Extra Axial Branch

GPT-5*

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Abstract

We locate ordinary Einstein–Maxwell gravitational waves inside pure Weyl–Maxwell gravity on the compact product $\mathbb{R}_t \times S_L^1 \times S^2$. The background is a common exact solution of both theories: the sphere has unit curvature, the Maxwell field is a unit parallel magnetic flux, and the action normalizations are $\kappa = 1$, $\Lambda = 1/2$, and $\alpha_B = 3$. On the fixed compact $U(1)$ bundle we classify the complete standard linearized Einstein–Maxwell harmonic phase space, including all axial and polar radiative modes, the physical exceptional $\ell = 1$ quotient, the six-dimensional homogeneous generalized block, and the three axial twist pairs.

The identity tangent map into Weyl–Maxwell theory is injective and its Lee–Wald pullback is nondegenerate on every standard block. It is not a symplectic inclusion. For $\ell \geq 2$ the target form is the Einstein–Maxwell form composed with the parity-independent spectral polynomial

$$p_\lambda(M) = 1 + \frac{3}{2}(M - \lambda),$$

whose two normal-mode weights are $1 \pm \frac{3}{2}\sqrt{2\lambda}$. Thus each real spatial harmonic has relative signature $(2, 2)$. The physical $\ell = 1$ block instead acquires the factor 4, the homogeneous block a nontrivial unipotent shear, and every axial twist pair the factor -2 .

We then solve the complete generic axial Weyl–Maxwell target. Modulo the Einstein image, it contains two additional cyclic summands on $\omega^2 - k^2 = \lambda - 2/3$. Their direct four-dimensional Lee–Wald Gram matrix is nondegenerate and positive in the declared positive-frequency current convention. The Einstein and extra modules are mutually orthogonal, and the complete generic axial target has classical signature $(3, 1)$. Inverting the extra Gram matrix gives two conserved coefficient detectors which annihilate the Einstein image.

These are classical LOCAL-ALGEBRAIC/REDUCED-MODE statements before the final residual quotient. They prove neither a positive-frequency Hilbert space nor a quantum ghost theorem. They also do not construct asymptotically flat scattering. The ordinary waves are present and nonnull before the global residual reduction; a vanishing one-particle residual cohomology on a closed cylinder therefore answers a different question from the existence of local gravitational radiation.

Authorship and computational inputs

The named model author synthesized the exact results and wrote the manuscript. The mathematical inputs are independently generated exact certificates in the project repository. Section 9 gives their

*OpenAI model. Contributed the mathematical synthesis, claim audit, and manuscript. The project was commissioned and directed by Asger Alstrup Palm (asger@area9.dk), who initiated the questions, coordinated the programme, and serves as corresponding human contact.

content hashes and verification commands. No floating-point calculation is used in a rank, signature, quotient, or on-shell identity reported here.

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1 The question and the answer

Every four-dimensional Einstein metric is Bach-flat, but the converse is false: conformal gravity has genuine non-Einstein solution branches [1, 5, 6]. In the coupled theory the relation is subtler still, because the Maxwell stress appears in both equations with different gravitational operators. The relevant question is consequently not whether one can fix Weyl gauge to obtain Einstein gravity. One cannot do so in general. It is instead:

On a common Einstein–Maxwell/Weyl–Maxwell background, which ordinary wave solutions embed, what phase-space form do they inherit, and which additional Weyl–Maxwell solutions remain?

This paper answers that question completely for the standard harmonic Einstein–Maxwell tangent and for the generic axial part of the additional target on a fixed compact magnetic bundle. The answer has three logically separate layers:

1. every standard Einstein–Maxwell tangent class injects as a Weyl–Maxwell solution class;
2. the target Lee–Wald form restricts nondegenerately to that image, but does not equal the source form under the identity map; and
3. the generic axial Weyl–Maxwell target contains two further nonradical solution directions which are not Einstein representatives or local gauge.

Let $\mathcal{T}_{\text{EM}}^{\text{std}}$ denote the complete fixed-bundle standard Einstein–Maxwell harmonic tangent after local $\text{Diff} \times U(1)$ reduction, and let $\mathcal{T}_{\text{WM}}^{\text{ax}}$ denote the generic axial Weyl–Maxwell tangent after local $\text{Diff} \times \text{Weyl} \times U(1)$ reduction. The central result can be stated schematically as

$$\boxed{\begin{aligned} \iota : \mathcal{T}_{\text{EM}}^{\text{std}} &\hookrightarrow \mathcal{T}_{\text{WM}}, & \ker(\iota^* \Omega_{\text{WM}}) &= 0, \\ & & \iota^* \Omega_{\text{WM}} &\neq \Omega_{\text{EM}}, \\ \mathcal{T}_{\text{WM}}^{\text{ax}} / \iota(\mathcal{T}_{\text{EM}}^{\text{ax}}) &\cong (F[\omega]/(p))^2, & p &= \omega^2 - k^2 - \lambda + \frac{2}{3}. \end{aligned}} \quad (1)$$

Here $F = \text{Frac} \mathbb{R}(\lambda, k)$ on the generic stratum. The last line is a canonical module quotient, not a choice of complement.

1.1 What is and is not frozen

The theorem level of this manuscript is deliberately asymmetric in parity. The standard Einstein–Maxwell image is complete in both axial and polar parities and includes all exceptional/global blocks. The additional Weyl–Maxwell branch is complete only in the generic axial sector. The polar extra branch and the final residual descent remain open. Accordingly the paper makes no parity-complete claim about the full Weyl–Maxwell target.

The dependency labels are

$$\text{LOCAL-ALGEBRAIC} \quad \text{and} \quad \text{REDUCED-MODE}.$$

No result below carries a LORENTZIAN-CAUSAL dependency label. The compact background is Lorentzian and the currents are conserved in real time, but no full Green-hyperbolic BV complex, Hadamard state, null-infinity phase space, or scattering construction is used.

2 The two theories and their common compact background

We use signature $(-, +, +, +)$ and the actions

$$S_{\text{EM}}[g, A] = \int_M \sqrt{-g} \left[\frac{R - 2\Lambda}{2\kappa} - \frac{1}{4} F_{ab} F^{ab} \right] d^4x, \quad (2)$$

$$S_{\text{WM}}[g, A] = \int_M \sqrt{-g} \left[\frac{\alpha_B}{8} C_{abcd} C^{abcd} - \frac{1}{4} F_{ab} F^{ab} \right] d^4x. \quad (3)$$

Thus

$$G_{ab} + \Lambda g_{ab} = \kappa T_{ab}, \quad \nabla_a F^{ab} = 0, \quad (4)$$

$$\alpha_B B_{ab} = T_{ab}, \quad \nabla_a F^{ab} = 0, \quad (5)$$

with $dF = 0$ and

$$T_{ab} = F_{ac} F_b{}^c - \frac{1}{4} g_{ab} F_{cd} F^{cd}. \quad (6)$$

Consider a product $M_2(k_1) \times \Sigma_2(k_2)$ of oriented constant curvature surfaces and an aligned parallel field

$$F = E \operatorname{vol}(M_2) + P \operatorname{vol}(\Sigma_2). \quad (7)$$

The same metric and field solve both theories on the incidence branch

$$\Lambda = \frac{k_1 + k_2}{2}, \quad E^2 + P^2 = \frac{k_2 - k_1}{\kappa}, \quad \alpha_B \kappa (k_1 + k_2) = 3. \quad (8)$$

This is an intersection of solution loci, not a gauge equivalence.

We specialize to the positive flat branch

$$M = \mathbb{R}_t \times S_L^1 \times S^2, \quad (k_1, k_2) = (0, 1), \quad (\kappa, \Lambda, \alpha_B, E, P) = (1, \tfrac{1}{2}, 3, 0, 1). \quad (9)$$

The spatial Cauchy surface is the closed manifold $S^1 \times S^2$. With minimal charge $q_{\min} = 1$, flux quantization gives Chern number $N = 2$. Throughout the paper the bundle P_N is fixed. Uniform magnetic variation would change the topological sector and is not an allowed tangent.

2.1 Why the complete linear inclusion holds

For an Einstein–Maxwell solution, the difference between the Bach equation and the Einstein equation can be expressed through the Chevreton tensor of the Maxwell field. On the product fixture the background flux is parallel. The Chevreton defect [2] is quadratic in ∇F , so both its background value and first variation vanish. With the coupling relation (8), every complete linearized Einstein–Maxwell solution therefore satisfies the complete linearized Weyl–Maxwell equations with the same pair (h, a) .

This is an on-shell tangent statement. It is stronger than principal-symbol inclusion and weaker than an off-shell BV chain map. It also predicts its own limitation: the first possible Chevreton failure is quadratic, so no nonlinear closure follows from the linear theorem.

Let

$$\iota[(h, a)]_{\text{Diff} \times U(1)} = [(h, a)]_{\text{Diff} \times \text{Weyl} \times U(1)}. \quad (10)$$

The map is well defined on the declared linear solution quotients. It is also injective. Indeed, if an Einstein class becomes target gauge, then after subtracting its common diffeomorphism and $U(1)$ part it has the form $(h_{ab}, a_a) = (2\sigma \bar{g}_{ab}, 0)$. The linearized Einstein–Maxwell rows imply

$$-3\Delta_{S^2}\sigma + 2\sigma = 0. \quad (11)$$

On $Y_{\ell m}$ this is multiplication by $3\ell(\ell + 1) + 2$, hence smoothness forces $\sigma = 0$.

Proposition 2.1 (Linear quotient inclusion). *On the fixed background (9), the identity tangent map sends every complete linearized Einstein–Maxwell solution to a linearized Weyl–Maxwell solution and descends injectively through the respective local gauge quotients.*

Remark 2.2. Proposition 2.1 does not say that the two theories are gauge equivalent. In particular, a generic Bach-flat solution need not be conformally Einstein, and Section 5 constructs additional target directions explicitly.

3 The complete standard Einstein–Maxwell harmonic phase space

Write

$$\lambda = \ell(\ell + 1), \quad k = k_n = \frac{2\pi n}{L}, \quad -\Delta_{S^2} Y_{\ell m} = \lambda Y_{\ell m}, \quad (12)$$

and $N_{\ell m} = \int_{S^2} |Y_{\ell m}|^2 d\Omega > 0$. A complex coefficient at (n, ℓ, m) is paired by reality with the coefficient at $(-n, \ell, -m)$. Equivalently one may use real spherical harmonics and cosine/sine circle modes. We use the latter convention for real mode counts.

3.1 Regular radiative blocks

For $\ell \geq 2$, the axial and polar reductions each have two masters. In the orders (H, Q) and (K, U) their normal-mode matrices are

$$M_A = \begin{pmatrix} \lambda & 2 \\ \lambda & \lambda \end{pmatrix}, \quad M_P = \begin{pmatrix} \lambda & -2\lambda \\ -1 & \lambda \end{pmatrix}. \quad (13)$$

Both parities have the exact dispersions

$$\omega_{\pm}^2 = k^2 + \lambda \pm \sqrt{2\lambda}. \quad (14)$$

The source action-normalized Wronskians are controlled, up to the common positive factor $N_{\ell m}/2$, by

$$G_A = \begin{pmatrix} \lambda^2 & 2\lambda \\ 2\lambda & 2\lambda \end{pmatrix}, \quad G_P = \begin{pmatrix} 1 & -2 \\ -2 & 2\lambda \end{pmatrix}. \quad (15)$$

They obey $G_A M_A = M_A^T G_A$ and $G_P M_P = M_P^T G_P$. Their determinants are $2\lambda^2(\lambda - 2)$ and $2(\lambda - 2)$, so both are positive for $\lambda \geq 6$. This is positivity of the classical source kinetic branch weights, not yet a one-particle Hilbert norm.

The branch eigenvectors may be taken as

$$v_{A,\pm} = (1, \pm\sqrt{\lambda/2}), \quad v_{P,\pm} = (\mp\sqrt{2\lambda}, 1). \quad (16)$$

These compact parity masters are the exact representation of the ordinary coupled photon/graviton-like waves. On the magnetically supported product the metric and Maxwell coefficients mix; helicity ± 2 is therefore a local/high-frequency interpretation, whereas (n, ℓ, m) , parity, and branch are the exact global labels.

3.2 Exceptional and global blocks

At $\ell = 1$ one branch becomes gauge in each parity. The axial source matrix has kernel $(H, Q) = (1, -1)$ and physical direction $(1, 1)$. The polar source matrix has kernel $(K, U) = (2, 1)$ and quotient coordinate $\Psi_P = U - K/2$. The surviving axial and polar modes obey $\omega^2 = k^2 + 4$. They are radiative triplets, not the zero-frequency twist.

The complete homogeneous generalized block is six-dimensional. A convenient representative is

$$K = a + bt, \quad C = at^2 + \frac{b}{3}t^3 + c + dt, \quad A_x = W_x + Q_e t. \quad (17)$$

Polynomial Jordan partners are retained; imposing boundedness in time would define a different phase space. After the common factor $2\pi L$ is removed, the source form in the order (a, b, c, d, Q_e, W_x)

is

$$\Omega_{\text{EM}}^{(0)} = \begin{pmatrix} 0 & -1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 & 2 & 0 \end{pmatrix}. \quad (18)$$

It has rank six. The flat holonomy W_x is the canonical partner of the electric charge Q_e and must not be deleted merely because it is invisible to the field strength.

For every one of the three real axial $\ell = 1$ harmonics, the generalized twist

$$h_{xa} = (A_m + B_m t)X_a, \quad a_x = -(A_m + B_m t)Y_{1m} \quad (19)$$

forms a Darboux pair. The would-be gauge generator is proportional to $x(A_m + B_m t)$ and is not periodic on S^1 .

Theorem 3.1 (Complete source phase space). *Before the final residual quotient, the fixed-bundle standard harmonic Einstein–Maxwell tangent is the symplectic direct sum*

$$\mathcal{T}_{\text{EM}}^{\text{std}} = \mathcal{T}_{\text{rad}}^{\ell \geq 2} \oplus \mathcal{T}_{\text{phys}}^{\ell=1} \oplus \mathcal{T}_{\text{hom}}^{\ell=0} \oplus \mathcal{T}_{\text{twist}}^{\ell=1, \omega=0}. \quad (20)$$

The regular radiative forms, exceptional physical quotient, homogeneous block, and three twist pairs are all nondegenerate.

4 The Weyl–Maxwell pullback on the Einstein image

A solution inclusion need not preserve a symplectic structure. We use the covariant phase-space current in the Lee–Wald convention [3, 4] and therefore keep three objects separate:

$$\Omega_{\text{EM}}, \quad \iota^* \Omega_{\text{WM}}, \quad \Omega_{\text{WM}} \text{ on the full target.} \quad (21)$$

The first two live on the same vector space but arise from different actions.

4.1 Regular radiation: one spectral polynomial

On both parities and for every $\ell \geq 2$,

$$\iota^* \Omega_{\text{WM}}(u, v) = \Omega_{\text{EM}}(u, p_\lambda(M)v), \quad p_\lambda(x) = 1 + \frac{3}{2}(x - \lambda). \quad (22)$$

In the axial and polar master coordinates,

$$p_\lambda(M_A) = \begin{pmatrix} 1 & 3 \\ 3\lambda/2 & 1 \end{pmatrix}, \quad p_\lambda(M_P) = \begin{pmatrix} 1 & -3\lambda \\ -3/2 & 1 \end{pmatrix}. \quad (23)$$

On the normal branches its eigenvalues are

$$r_+ = 1 + \frac{3}{2}\sqrt{2\lambda} > 0, \quad r_- = 1 - \frac{3}{2}\sqrt{2\lambda} < 0. \quad (24)$$

Neither vanishes for $\lambda \geq 6$. The axial and polar branch weights are identical even though their off-shell coefficient matrices differ.

For a real spatial harmonic there are four oscillator blocks: two parities times two branches. Hence the relative coefficient signature is

$$\text{sig}_{\text{rel}}(\iota^* \Omega_{\text{WM}}) = (2, 2). \quad (25)$$

This is shorthand for the signs of the symmetric branch coefficients that multiply the standard real Darboux blocks. A symplectic form itself has no ordinary Sylvester signature.

4.2 Exceptional and global restrictions

The exceptional $\ell = 1$ calculation cannot be obtained by blindly setting $\lambda = 2$ in the generic polar matrix: that continuation fails target gauge descent. Direct four-dimensional currents instead give, in source-normalized axial and polar quotient coordinates,

$$\iota^* \Omega_{\text{WM}}|_{\ell=1, \text{phys}} = 4 \Omega_{\text{EM}}|_{\ell=1, \text{phys}}. \quad (26)$$

The literal target current has zero gauge rows and columns, so this is a quotient identity rather than only a norm comparison.

On the homogeneous block, after the same common factor as in (18),

$$\Omega_{\text{WM}}^{(0)} = \begin{pmatrix} 0 & 2 & 0 & -1 & 0 & 0 \\ -2 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 & 2 & 0 \end{pmatrix}. \quad (27)$$

Writing $R = (\Omega_{\text{EM}}^{(0)})^{-1} \Omega_{\text{WM}}^{(0)}$, one obtains

$$R = I + N, \quad N^2 = 0, \quad \text{rank } N = 2. \quad (28)$$

Thus the identity is not symplectic, although $S = I + N/2$ has determinant one and satisfies $S^T \Omega_{\text{EM}}^{(0)} S = \Omega_{\text{WM}}^{(0)}$.

Finally, each generalized axial twist pair obeys the direct identity

$$\iota^* \Omega_{\text{WM}}|_{\text{twist}} = -2 \Omega_{\text{EM}}|_{\text{twist}}. \quad (29)$$

It comes from the full $A + Bt$ current and is not a zero-frequency limit of an oscillatory formula.

4.3 Orthogonality and the assembled theorem

Different circle momenta, spherical irreducibles, and parity sectors are orthogonal by S^1 Fourier invariance, $SO(3)$ invariance, and spatial parity. Distinct radiative branches are orthogonal because M is Ω_{EM} -self-adjoint and has distinct eigenvalues. The only shared label not settled by these arguments is the physical axial $\ell = 1, n = 0$ oscillator against the twist. Its literal mixed target current is proportional to

$$-2i\pi\omega p(\omega^2 - 4)[\omega(A + Bt) - iB]e^{-i\omega t}, \quad (30)$$

and therefore vanishes on the physical shell $\omega^2 = 4$ for both Jordan coordinates.

Theorem 4.1 (Complete standard harmonic inclusion). *On the fixed compact background (9), and before the final residual quotient, the identity map*

$$\iota : \mathcal{T}_{\text{EM}}^{\text{std}} \hookrightarrow \mathcal{T}_{\text{WM}} \quad (31)$$

has nondegenerate target pullback:

$$\ker \left(\iota^* \Omega_{\text{WM}}|_{\mathcal{T}_{\text{EM}}^{\text{std}}} \right) = 0. \quad (32)$$

The pullback is the block-orthogonal direct sum of (22), (26), (27), and (29). The identity inclusion is not symplectic.

Corollary 4.2 (Restricted linear observables). *On the restricted Einstein image, every linear Einstein–Maxwell observable has a unique target-restricted Hamiltonian vector after application of the inverse blockwise relative operator.*

Remark 4.3. The corollary is not an embedding into the full target observable algebra. Extending an observable away from the Einstein image and descending it under the final residual action are separate questions.

5 The complete generic axial extra branch

We now solve the fourth-order target rather than restricting it to the Einstein image. For generic $\ell \geq 2$, after local gauge contraction the axial coefficient order is

$$\Phi = (H_t, H_x, Q_t, Q_x)^T. \quad (33)$$

The contraction inverts only the harmless constant 2: neither k nor ω is inverted, so zero momentum is retained.

Put $s = \omega^2 - k^2$. The Einstein master polynomial and the extra polynomial are

$$q(s) = (s - \lambda)^2 - 2\lambda, \quad p(s) = s - \lambda + \frac{2}{3}. \quad (34)$$

Their resultant vanishes only at the algebraic collision $\lambda = 2/9$, which is disjoint from the physical range $\lambda \geq 6$. The exact Smith/module reduction gives

$$\mathcal{Q}_{\text{ax}}^{\text{extra}} := \mathcal{T}_{\text{WM}}^{\text{ax}} / \iota(\mathcal{T}_{\text{EM}}^{\text{ax}}) \cong (F[\omega]/(p))^2, \quad F = \text{Frac } \mathbb{R}(\lambda, k). \quad (35)$$

Thus there are two independent additional cyclic summands on

$$\omega_e^2 = k^2 + \lambda - \frac{2}{3}. \quad (36)$$

Convenient representatives are

$$e_1 = \begin{pmatrix} -k^2 - \lambda \\ k\omega_e \\ \lambda \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} -k\omega_e \\ k^2 - \frac{2}{3} \\ 0 \\ \lambda \end{pmatrix}. \quad (37)$$

They are not defined by an arbitrary orthogonal complement: they represent a canonical quotient supported on a different invariant factor.

5.1 Direct Lee–Wald pairing

Normalize the direct current by $J^t/(-i\omega_e N_{\ell m})$ at positive frequency, suppressing the common positive circle volume. In the basis (37) its Gram matrix G_X has entries

$$(G_X)_{11} = \frac{\lambda}{6} [9k^2\lambda^2 - 12k^2\lambda + 4k^2 + 9\lambda^3 - 18\lambda^2], \quad (38)$$

$$(G_X)_{12} = \frac{k\lambda\omega_e}{6} (3\lambda - 2)^2, \quad (39)$$

$$(G_X)_{22} = \frac{\lambda}{18} [27k^2\lambda^2 - 36k^2\lambda + 12k^2 + 36\lambda - 8]. \quad (40)$$

On the extra shell,

$$\det G_X = \frac{1}{3} \lambda^4 (\lambda - 2)(9\lambda - 2) > 0. \quad (41)$$

The first principal minor is also positive for every real k and $\lambda \geq 6$. Hence $\text{sig } G_X = (2, 0)$.

The Einstein and extra primary modules are mutually Lee–Wald orthogonal. The two Einstein master branches contribute one positive and one negative coefficient in the same direct-current convention. Therefore:

Theorem 5.1 (Generic axial target classification). *For every $\ell \geq 2$, every real compact momentum k , and before the final residual quotient, the complete generic axial Weyl–Maxwell solution module is the Lee–Wald orthogonal direct sum of the two Einstein master branches and the rank-two extra module (35). It is nondegenerate and has classical coefficient signature*

$$\text{sig } \mathcal{T}_{\text{WM}}^{\text{ax}} = (3, 1). \quad (42)$$

The extra block has signature (2, 0); the negative target direction lies on the minus Einstein-image branch.

Remark 5.2 (What the sign does not prove). Theorem 5.1 is a sign theorem for a classical conserved current after a declared positive-frequency normalization. No compatible complex structure, positive one-particle inner product, BRST physical Hilbert space, or quantum time evolution has been constructed. Consequently the signature (3, 1) is not, by itself, a quantum ghost or unitarity theorem. The negative target coefficient belongs to $\iota^* \Omega_{\text{WM}}$, not to the independent Einstein–Maxwell source action. Familiar higher-derivative ghost conclusions [7] require additional quantum structure not constructed here.

5.2 A detector for the extra coefficients

Nondegeneracy makes the extra branch measurable within the declared reduced solution block. For an on-shell axial solution Φ , define

$$\mathcal{O}_X(\Phi) = G_X^{-1} \frac{\Omega_{\text{WM}}(\bar{e}, \Phi)}{-i\omega_e N_{\ell m}} \in \mathbb{C}^2, \quad (43)$$

where $e = (e_1, e_2)^T$. The local Green identity gives slice independence, and direct substitution gives

$$\mathcal{O}_X(a_1 e_1 + a_2 e_2) = (a_1, a_2)^T, \quad \mathcal{O}_X \circ \iota = 0 \quad (44)$$

on the complete generic axial Einstein image. Thus the additional solutions are neither a reduced-current radical nor a relabeling of Einstein modes. The detector has not yet been descended through the final residual action or promoted to a relational, causal, asymptotic, or quantum observable.

6 The normalization triangle

Higher-derivative currents are sensitive to integration-by-parts and action normalization conventions. Three independent constructions are therefore compared. First, the gauge-reduced axial equation operator is formally self-adjoint. In Fourier convention $\partial_t \mapsto -i\omega$, $\partial_x \mapsto ik$, its exact quadratic kernel in the order (H_t, H_x, Q_t, Q_x) is

$$K = \begin{pmatrix} -\frac{\lambda}{4}A & -\frac{k\lambda\omega}{4}B & \lambda & 0 \\ -\frac{k\lambda\omega}{4}B & \frac{\lambda}{4}C & 0 & -\lambda \\ \lambda & 0 & k^2 + \lambda & k\omega \\ 0 & -\lambda & k\omega & -\lambda + \omega^2 \end{pmatrix}, \quad (45)$$

where

$$A = 3k^4 + 6k^2\lambda - 3k^2\omega^2 - 2k^2 + 3\lambda^2 - 3\lambda\omega^2 - 8\lambda + 6\omega^2, \quad (46)$$

$$B = 3k^2 + 3\lambda - 3\omega^2 + 4, \quad (47)$$

$$C = 3k^2\lambda - 3k^2\omega^2 - 6k^2 + 3\lambda^2 - 6\lambda\omega^2 - 8\lambda + 3\omega^4 + 2\omega^2. \quad (48)$$

It satisfies $K(\omega, k)^\dagger = K(-\omega, -k)^T$ and is the mixed Hessian of

$$S_{\text{red}}^{(2)} = \frac{1}{2} \int \Phi(-\omega, -k)^T K(\omega, k) \Phi(\omega, k). \quad (49)$$

Second, the Green current generated by this Hessian obeys the exact local Green identity. Third, independent direct variation of the four-dimensional Weyl–Maxwell action gives a Lee–Wald current whose integral over S^2 equals $N_{\ell m}$ times that reduced Green current. Complete independent-frequency matrices were checked at $\ell = 2, 3, 4$; natural $SO(3)$ equivariance and the degree-two spectral bound promote the equality to every $\ell \geq 2$ and every m . No current improvement remains after compact-sphere integration.

Proposition 6.1 (Hessian–Green–Lee–Wald triangle). *In the generic axial block, the exact equation operator is the Hessian of the reconstructed quadratic reduced action; its local Green current equals the direct integrated four-dimensional Lee–Wald current with the normalization used in Theorem 5.1.*

Proposition 6.1 is sufficient for the linear phase-space and signature theorems: it fixes the operator, local conservation law, direct four-dimensional current, and overall action sign independently. A literal second expansion of the unreduced four-dimensional action density would be a valuable additional audit, but it is not used or claimed here.

7 Where did the graviton go?

The answer depends on which quotient and which boundary problem is being asked about.

Before the final residual quotient. The usual gravitational radiation is represented by the axial and polar metric-containing normal modes (14), together with their flux-forced Maxwell dressings. Every one of these standard modes has a nonzero source Lee–Wald pairing and a nonzero target pullback pairing. The physical $\ell = 1$ quotient, generalized homogeneous modes, and twist pairs are nondegenerate as well. The graviton-like tangent has therefore not been removed at this stage.

After a global residual quotient. A residual cohomology calculation may further identify states under a global conformal action on a closed compact universe. If its one-particle sector vanishes, that statement concerns invariants of the fully reduced global state space. It does not retroactively make the pre-quotient solutions zero, nor does it show that local wave packets cannot solve the equations. The coefficient detectors (43) likewise exist before this last descent and may or may not survive it.

Closed cylinder versus scattering. The Cauchy surface here is $S^1 \times S^2$ and has no null infinity. Asymptotically flat gravitational radiation is organized by Bondi news, symplectic flux through $\mathcal{I}^\pm$, and charged asymptotic symmetries. Transformations that are proper gauge on a compact slice can acquire boundary charges and cease to be gauge in that setting. Consequently a compact residual-cohomology statement cannot be used as evidence that the ordinary helicity- ± 2 scattering space vanishes.

Einstein is a sector, not a Weyl gauge. The exact relation established here is

$$\text{Sol}_{\text{EM}}^{\text{lin, std}} \subsetneq \text{Sol}_{\text{WM}}^{\text{lin, ax+standard}} \quad (50)$$

on the declared background and harmonic strata. The strictness is witnessed by (35). Boundary conditions may later select an Einstein sector, but such selection is additional dynamics plus boundary data, not a local Weyl gauge fixing.

8 Open gates and theorem boundary

The following are not prerequisites for the scoped axial paper, but they are required to widen its title-level theorem.

1. **Polar extra branch.** Repeat the exact operator/module, Lee–Wald, radical, sign, and detector analysis in even parity. Only then is the complete additional compact target parity-classified.
2. **Final residual descent.** Determine which standard and extra coefficient observables survive the declared global residual quotient. Until then physical mode means the local-gauge-reduced compact phase space, not the final residual state space.
3. **Literal action-density expansion.** This is an independent normalization audit. The present theorem instead uses the exact Hessian–Green–direct–Lee–Wald triangle of Proposition 6.1.
4. **Nonlinear extension.** Linear inclusion does not imply that an Einstein tangent extends to a Weyl–Maxwell family. Fixed-bundle Taub tests already show scoped second-order obstructions; those belong to a separate nonlinear charge-fibre paper.
5. **Causal boundary selection and scattering.** A future theorem must specify asymptotically flat function spaces, retarded/advanced complexes, radiative symplectic flux, residual charges, and whether the Einstein subspace is dynamically closed. Bondi energy, black holes, and scattering are not inferred here.

The strongest present conclusion is therefore:

On the fixed compact product, ordinary Einstein–Maxwell waves embed exactly and nondegenerately in the linear Weyl–Maxwell phase space, but with a different Hamiltonian form. Weyl–Maxwell theory also has a complete generic axial branch of genuine, nonradical additional solutions. Neither branch has yet been classified after the final residual quotient or as an asymptotic quantum particle space.

9 Exact certification and reproducibility

Every theorem in this manuscript is a synthesis of exact machine-readable certificates. The companion claim map `paper/10-compact-einstein-maxwell-weyl-phase-space-claim-map.json` pins the full SHA-256 hashes and fails closed if an input changes.

Result	Certificate role
EINSTEIN_MAXWELL_PRODUCT_INCIDENT	Common exact product background and fixed-flux rational fixture.
EINSTEIN_MAXWELL_CHEVRETON_TANGENT	Complete linear on-shell solution inclusion.
COMPACT_EM_RADIATIVE_SYMPLECTIC_MATCHING	Positive source radiative pairing and direct Lee–Wald match.
EINSTEIN_MAXWELL_WEYL_STANDARD_HARMONIC_SYMPLECTIC_INCLUSION	Complete standard image, mixed orthogonality, and nondegenerate pullback.
EINSTEIN_MAXWELL_WEYL_AXIAL_OPERATOR	Generic target operator, invariant factors, and extra quotient.
EINSTEIN_MAXWELL_WEYL_AXIAL_LEE_WALD_COMPLETION	Direct four-dimensional current and complete axial signature.
EINSTEIN_MAXWELL_WEYL_AXIAL_REDUCTION_ACTION_HESSIAN	Reduced Hessian and normalization triangle.
EINSTEIN_MAXWELL_WEYL_AXIAL_EXTRA_DETECTOR	Conserved extra coefficient observables before residual descent.

Table 1: Principal exact inputs. The claim map also pins the exceptional and mixed-block component certificates.

The scoped verification commands are listed below; the breakable path formatting is typographical and each line is one shell command.

```
python3bridge/einstein_sector/verify_einstein_maxwell_product_incidence.py
python3bridge/einstein_sector/verify_einstein_maxwell_chevreton_tangent.py
python3bridge/einstein_sector/verify_einstein_maxwell_radiative_symplectic_matching.py
python3bridge/einstein_sector/verify_einstein_maxwell_weyl_standard_harmonic_symplectic_inclusion.py
python3bridge/einstein_sector/verify_einstein_maxwell_weyl_axial_operator.py
python3bridge/einstein_sector/verify_einstein_maxwell_weyl_axial_lee_wald_completion.py
python3bridge/einstein_sector/verify_einstein_maxwell_weyl_axial_reduced_action_hessian.py
python3bridge/einstein_sector/verify_einstein_maxwell_weyl_axial_extra_detector.py
python3bridge/einstein_sector/verify_compact_linear_paper_claim_map.py
```

The manuscript itself is checked with two passes of `pdflatex`.

10 Conclusion

The Einstein sector is visible without interpretive guesswork. On a common compact Einstein–Maxwell/Weyl–Maxwell background, the complete standard Einstein–Maxwell harmonic tangent injects into the Weyl–Maxwell solution space and remains nondegenerate under the target Lee–Wald form. The identity inclusion nevertheless changes that form in every class of blocks: by branch-

dependent weights on regular radiation, a factor four on the physical $\ell = 1$ quotient, a unipotent shear on the homogeneous sector, and a factor minus two on the twists.

The target is strictly larger. Its generic axial quotient contains two additional nonradical solution summands, detected by conserved coefficient observables, and the complete direct-current axial signature is $(3, 1)$. These statements settle where ordinary waves and one genuine extra branch sit before the final residual quotient. They do not turn a classical sign into a quantum ghost theorem, and they do not replace the separate causal problem of selecting an Einstein scattering sector at infinity.

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